Bright Coffee Shop Dashboard

Coffee Shop Sales Performance Dashboard

About This Dashboard

This dashboard provides a comprehensive analysis of coffee shop sales performance across three store locations. It tracks revenue, order volume, product performance, and spending patterns to support data-driven business decisions for the Bright Coffee Shop chain.

The dashboard is built on transaction-level data from the point-of-sale system, covering every individual purchase recorded across all store locations. It enables exploration by date range, product category, product type, store location, and individual product.

Data Refresh Information

Detail	Value
Data Source	`coffeeshop.brightlearn.brightlearn_coffee_shop_2`
Reporting Period	January 2023 – June 2023 (181 days)
Currency	South African Rand (ZAR)
Total Records	149,116 transactions
Store Locations	3 (Hell's Kitchen, Astoria, Lower Manhattan)

Dashboard Contents

Page	Description
Cover	This page — dashboard overview, data source details, and refresh information
Summary	Executive summary with highlighted KPIs, key filters, and performance highlights
Detailed View	Technical breakdown of how each metric is calculated, including formulas and data lineage
Sales Performance	Interactive dashboard with KPIs, filters, charts, and detailed data table
Glossary	Definitions of all terminology and abbreviations used in the dashboard

How to Use This Dashboard

1. **Navigate** between pages using the tabs at the top
2. **Apply filters** on the Sales Performance page to drill down by date range, product category, product type, store location, or specific products
3. **Hover** over any chart for detailed tooltips with exact values
4. **Sort** table columns by clicking column headers
5. **All KPIs and charts update automatically** when filters are applied

Key Metrics at a Glance

Metric	Value
Total Revenue	R 698,812
Total Orders	149,116
Average Order Value	R 4.69
Total Products	80 across 9 categories
Revenue Growth	104% (Jan → Jun 2023)

Executive Summary

Key Performance Indicators (KPIs)

The dashboard tracks **six core KPIs** that together provide a complete picture of business performance:

KPI	Value	What It Measures
Total Revenue	R 698,812	Sum of all sales: `unit_price × transaction_qty` across all transactions
Total Orders	149,116	Count of unique transactions (distinct `transaction_id`)
Total Units Sold	214,470	Sum of `transaction_qty` — total individual items sold
Avg Order Value	R 4.69	Revenue divided by total orders — average spend per transaction
Total Products	80	Count of distinct `product_id` values across 9 categories
Total Stores	3	Count of distinct store locations (Hell's Kitchen, Astoria, Lower Manhattan)

Available Filters

The dashboard includes **five key filters** on the Sales Performance page. All filters are multi-select and affect every widget simultaneously:

Filter	Field	Description
Date Range	`transaction_date`	Select a start and end date to limit data to a specific period
Product Category	`product_category`	Filter by high-level category (Coffee, Tea, Bakery, etc.)
Product Type	`product_type`	Filter by sub-category within a category (e.g., Latte, Espresso)
Store Location	`store_location`	Filter by specific store (Hell's Kitchen, Astoria, Lower Manhattan)
Product Detail	`product_detail`	Filter by specific product name (e.g., "Dark Chocolate Lg")

Charts on the Sales Performance Page

All charts are styled in the dashboard's pink theme and respond to every filter:

Revenue Charts

- **Total Revenue by Month (line)** — monthly revenue trend showing 104% growth from Jan to Jun 2023
- **Daily Revenue Trend (line)** — day-by-day revenue with a separate line for each store location
- **Revenu by Product Category (pie)** — revenue split across the 9 product categories
- **Total Revenue by Store Location (bar)** — revenue distribution across the 3 stores
- **Revenue: Weekday vs Weekend by Month (grouped bars)** — weekday vs weekend revenue for each month
- **Top 10 Products: Revenue vs Units Sold (dual-axis horizontal bars)** — the 10 highest-revenue products, each shown with revenue and units sold on two independent scales

Volume Charts

- **Revenue by Day of Week (bar)** — order volume per day, ordered Monday through Sunday to reveal weekly patterns
- **Product Sales Summary (table)** — product × category × store breakdown with revenue, units, and orders

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Measured Values Covered

Revenue Metrics

- **Total Revenue** — overall sales revenue across the full period
- **Revenue by Month** — monthly trend showing 104% growth from Jan to Jun 2023
- **Daily Revenue Trend** — day-by-day revenue, split by store location
- **Revenue by Product Category** — breakdown across 9 categories
- **Revenue by Store Location** — distribution across 3 stores
- **Weekday vs Weekend Revenue** — monthly weekday/weekend comparison
- **Top 10 Products by Revenue** — highest-earning individual products

Volume Metrics

- **Total Orders** — transaction count
- **Total Units Sold** — individual items sold
- **Order Performance by Day of Week** — busiest and slowest days (ordered Mon–Sun)
- **Top 10 Products by Units Sold** — shown alongside revenue in the combined Top 10 chart

Derived Metrics

- **Average Order Value** — revenue per transaction

Key Findings

Revenue Growth

- Revenue **doubled** from R 81,678 in January to R 166,486 in June — a **104% increase** over 6 months
- Strongest month-over-month growth occurred in March (+30%) and May (+32%)

Category Performance

- **Coffee** dominates at R 269,952 (**38.6%** of total revenue)
- **Tea** follows at R 196,406 (**28.1%**)
- Together, Coffee and Tea account for **66.7%** of all sales
- **Underperformers:** Packaged Chocolate (R 4,408) and Flavours (R 8,409) contribute less than 2% combined

Regional Distribution

- Sales are **evenly distributed** across all three stores:
  - Hell's Kitchen: R 236,511 (33.8%)
  - Astoria: R 232,244 (33.2%)
  - Lower Manhattan: R 230,057 (32.9%)

Top Products

- **By Revenue:** Sustainably Grown Organic Lg (R 21,552), Dark Chocolate Lg (R 21,006), Latte Rg (R 19,112)
- **By Volume:** Earl Grey Rg (4,708 units), Dark Chocolate Lg (4,668 units), Morning Sunrise Chai Rg (4,643 units)
- Large-size drinks generate more revenue per unit; regular-size drives volume

Order Patterns

- **Busiest day:** Friday (21,701 orders)
- **Slowest day:** Saturday (20,510 orders)
- Weekday volume consistently exceeds weekends by ~3%

Recommendations

1. **Leverage Coffee & Tea dominance** — ensure consistent stock and quality for these primary revenue drivers
2. **Investigate underperforming categories** — Packaged Chocolate and Flavours may benefit from promotions or menu review
3. **Capitalise on weekend gap** — Saturday's lower volume suggests opportunity for weekend promotions or loyalty incentives
4. **Maintain balanced store performance** — share best practices across all three locations
5. **Monitor growth trajectory** — track the 104% revenue growth monthly to identify seasonal patterns

Detailed Metric Breakdown

This page provides a technical explanation of how each metric in the dashboard is calculated, including the underlying SQL formulas, data sources, and transformation logic.

Primary Dataset

All core KPIs are derived from a **local metric view** built on the source table:

Source Table: `coffeeshop.brightlearn.brightlearn_coffee_shop_2`

The metric view defines measures (pre-aggregated values) and dimensions (grouping fields) that power every widget on the dashboard. Each measure is computed using a specific SQL expression applied to the raw transaction data.

KPI Calculations

1. Total Revenue

Formula: `SUM(CAST(REPLACE(unit_price, ',', '.') AS DOUBLE) * transaction_qty)`

How it works:

- The `unit_price` column is stored as a **string** with European decimal format (comma separators, e.g., `"3,00"`)
- `REPLACE(unit_price, ',', '.')` converts the comma to a decimal point (e.g., `"3.00"`)
- `CAST(... AS DOUBLE)` converts the string to a numeric type
- The price is multiplied by `transaction_qty` (units purchased) for each transaction
- `SUM(...)` adds up the revenue across all transactions
- **Result:** R 698,812
- **Formatted as:** ZAR currency with 2 decimal places

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2. Total Orders

Formula: `COUNT(DISTINCT transaction_id)`

How it works:

- Each transaction has a unique `transaction_id`
- `COUNT(DISTINCT ...)` counts the number of unique transaction IDs
- **Result:** 149,116 orders

3. Total Units Sold

Formula: `SUM(transaction_qty)`

How it works:

- `transaction_qty` represents the number of items in each transaction (typically 1 or 2)
- `SUM(...)` adds up all quantities across all transactions
- This is distinct from Total Orders — one order can contain multiple units

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4. Average Order Value (AOV)

Formula: `MEASURE(revenue) / MEASURE(total_orders)`

How it works:

- Divides Total Revenue by Total Orders
- `MEASURE(revenue)` references the pre-defined revenue measure (calculation #1 above)
- `MEASURE(total_orders)` references the pre-defined order count (calculation #2 above)
- **Result:** R 698,812 ÷ 149,116 = R 4.69
- **Formatted as:** ZAR currency with 2 decimal places

5. Total Products

Formula: `COUNT(DISTINCT product_id)`

How it works:

- Each product has a unique `product_id`
- `COUNT(DISTINCT ...)` counts the number of unique product IDs
- **Result:** 80 distinct products across 9 categories

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6. Total Stores

Formula: `COUNT(DISTINCT store_id)`

How it works:

- Each store has a unique `store_id`
- `COUNT(DISTINCT ...)` counts the number of unique store IDs
- **Result:** 3 stores (Hell's Kitchen, Astoria, Lower Manhattan)

Chart Calculations

Revenue by Month (Line Chart)

Formula:

- **X-axis:** `DATE_TRUNC('MONTH', transaction_date)` — truncates the timestamp to the first day of the month
- **Y-axis:** `MEASURE(revenue)` — the revenue measure grouped by month
- Groups all transactions by month and sums the revenue for each

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Daily Revenue Trend (Line Chart)

Formula:

- **X-axis:** `DATE_TRUNC('DAY', transaction_date)` — truncates the timestamp to the date portion
- **Y-axis:** `MEASURE(revenue)` — the revenue measure grouped by day
- Groups all transactions by day and sums the revenue for each

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Revenue by Product Category (Bar Chart)

Formula:

- **Category axis:** `product_category` — groups transactions by category
- **Value axis:** `MEASURE(revenue)` — sums revenue per category
- Sorted descending by revenue to show highest-performing categories first

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Revenue by Store Location (Bar Chart)

Formula:

- **Category axis:** `store_location` — groups transactions by store
- **Value axis:** `MEASURE(revenue)` — sums revenue per store
- Sorted descending by revenue

Order Performance by Day of Week (Bar Chart)

Formula:

- **Category axis:** `DAYNAME(transaction_date)` — extracts the day name (Mon–Sun)
- **Value axis:** `MEASURE(total_orders)` — counts orders per day of week
- Groups all transactions by day name and counts distinct transactions

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Top 10 Products by Revenue (Horizontal Bar)

Formula:

- **Category axis:** `product_detail` — groups by full product name
- **Value axis:** `MEASURE(revenue)` — sums revenue per product
- Sorted descending by revenue, limited to top 10 visible categories (scrollable)

Top 10 Products by Quantity (Horizontal Bar)

Formula:

- **Category axis:** `product_detail` — groups by full product name
- **Value axis:** `MEASURE(total_units)` — sums quantity per product
- Sorted descending by quantity, limited to top 10 visible categories

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Product Sales Summary (Table)

Formula:

- Groups by `product_detail`, `product_category`, and `store_location`
- Columns: product name, category, store, `MEASURE(revenue)`, `MEASURE(total_units)`, `MEASURE(total_orders)`
- All three grouping dimensions combined, showing revenue/units/orders per product per store

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Secondary Dataset: Spending Tiers

A separate SQL dataset powers the Revenue by Spending Tier chart:

Spending Tier Classification:

```
CASE
  WHEN transaction_qty * CAST(REPLACE(unit_price, ',', '.') AS DOUBLE) <= 3.0  THEN 'Low'
  WHEN transaction_qty * CAST(REPLACE(unit_price, ',', '.') AS DOUBLE) <= 6.0  THEN 'Medium'
  ELSE 'High'
END
```

How it works:

- Each transaction's total spend is calculated as `quantity x unit_price`
- Transactions are classified into three tiers:
  - **Low:** Total spend ≤ R 3.00
  - **Medium:** Total spend ≤ R 6.00
  - **High:** Total spend > R 6.00
- The chart sums `transaction_spend` for each tier and displays as a bar chart

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Derived Dimensions

Dimension	Formula	Used In
`transaction_day`	`DATE(transaction_date)`	Daily Revenue Trend chart
`transaction_month`	`DATE_TRUNC('MONTH', transaction_date)`	Revenue by Month chart
`day_of_week`	`DAYNAME(transaction_date)`	Order Performance by Day of Week chart
`spending_tier`	CASE expression on `qty x price`	Revenue by Spending Tier chart

Unit Price Conversion

The `unit_price` field requires special handling because it is stored as a **string** with **comma decimal separators** (European format):

Stored Value	After REPLACE	After CAST	Example
`"3,00"`	`"3.00"`	`3.0`	R 3.00
`"2,50"`	`"2.50"`	`2.5`	R 2.50
`"4,75"`	`"4.75"`	`4.75`	R 4.75

This conversion is applied in both the revenue measure and the spending tier calculation.

Bright Coffee Shop Performance Dashboard

Total Revenue	Total Orders	Total Units Sold	Avg Order Value	Total Products	Total Stores
R 698,...	149.12k	214.47k	R 4.69	80	3

Date Range: mm/dd/yyyy --:--:-- — mm/dd/yyyy --:--:--

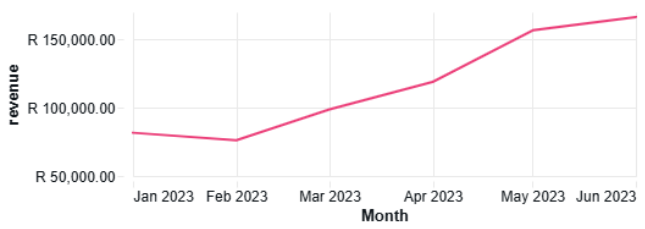
Product Category: All

Product Type: All

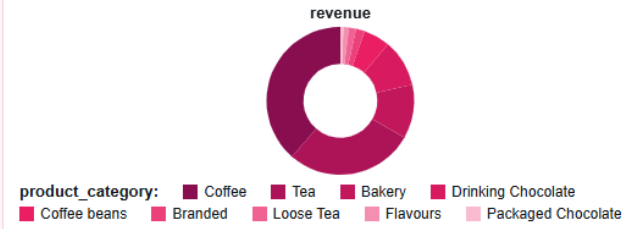
Store Location: All

Product Detail: All

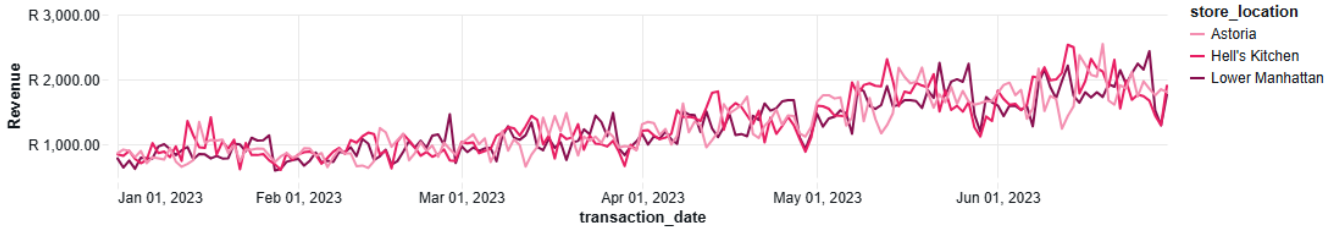
Total Revenue by Month



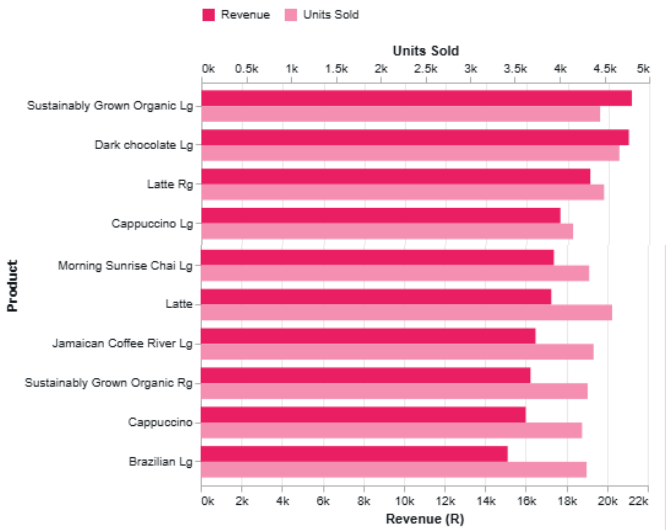
Revenu by Product Category



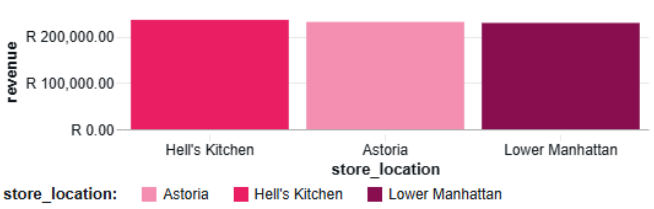
Daily Revenue Trend



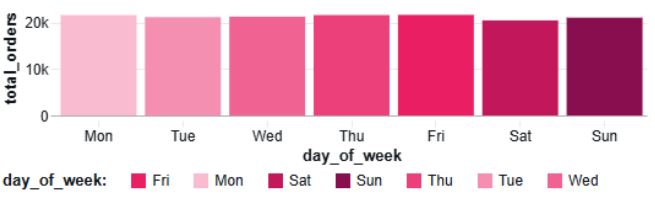
Top 10 Products: Revenue vs Units Sold



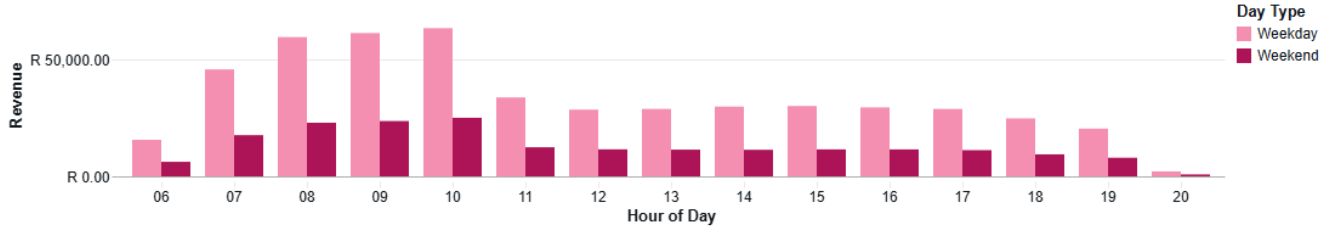
Total Revenue by Store Location



Revenue by Day of Week



Revenue by Time of Day



Product Sales Summary

product_detail	product_category	store_location	Revenue	Total Units	Total Orders
Ethiopia Rg	Coffee	Lower Manhattan	R 4,545.00	1515	930
Spicy Eye Opener Chai Lg	Tea	Lower Manhattan	R 4,231.50	1365	870
Dark chocolate Lg	Drinking Chocolate	Lower Manhattan	R 6,574.50	1461	912
Our Old Time Diner Blend Sm	Coffee	Lower Manhattan	R 2,718.00	1359	838
Oatmeal Scone	Bakery	Lower Manhattan	R 1,938.00	646	646
Columbian Medium Roast Sm	Coffee	Lower Manhattan	R 2,434.00	1217	773
Latte Rg	Coffee	Lower Manhattan	R 6,387.75	1503	923
Dark chocolate Rg	Drinking Chocolate	Lower Manhattan	R 4,308.50	1231	763

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Key Performance Insights

Key Performance Indicators

- **Total Revenue:** R 698,812 across 6 months (Jan–Jun 2023)
- **Total Orders:** 149,116 transactions processed
- **Avg Order Value:** R 4.69 per order
- **Total Products:** 80 distinct products across 9 categories
- **Total Stores:** 3 locations, each contributing ~33% of revenue

Business Insights

Top-Performing Category: Coffee leads with **R 269,952** in revenue (38.6% of total), followed by Tea at **R 196,406** (28.1%). Together, Coffee and Tea account for **66.7%** of all sales.

Sales Trend: Revenue grew **104%** from January (R 81,678) to June (R 166,486), with the strongest month-over-month growth in March (+30%) and May (+32%). The consistent upward trend indicates healthy business growth.

Business Insights

Top-Performing Category: Coffee leads with **R 269,952** in revenue (38.6% of total), followed by Tea at **R 196,406** (28.1%). Together, Coffee and Tea account for **66.7%** of all sales.

Sales Trend: Revenue grew **104%** from January (R 81,678) to June (R 166,486), with the strongest month-over-month growth in March (+30%) and May (+32%). The consistent upward trend indicates healthy business growth.

Regional Performance: Sales are remarkably evenly distributed across all three stores — Hell's Kitchen (R 236,511), Astoria (R 232,244), and Lower Manhattan (R 230,057) — each contributing roughly 33% of total revenue.

Top Products by Revenue: "Sustainably Grown Organic Lg" (R 21,552) and "Dark Chocolate Lg" (R 21,006) are the highest revenue-generating products. Premium large-size beverages dominate the top 5.

Top Products by Volume: "Earl Grey Rg" (4,708 units) and "Dark Chocolate Lg" (4,668 units) lead in units sold. Regular-size drinks drive volume, while large-size drinks generate more revenue per unit.

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Order Patterns: Friday is the busiest day (21,701 orders) and Saturday the slowest (20,510 orders). Weekday order volume consistently exceeds weekends by ~3%.

Underperforming Categories: Packaged Chocolate (R 4,408) and Flavours (R 8,409) contribute less than 2% of revenue combined. These may benefit from promotional strategies or menu repositioning.

Glossary

This page defines all terminology and abbreviations used throughout the Coffee Shop Sales Performance Dashboard.

Key Definitions

Term	Definition
Revenue	Total sales revenue. Calculated as `unit_price x transaction_qty` for each transaction, then summed. Displayed in ZAR.
Order	A single customer purchase event, identified by a unique `transaction_id`.
Unit	An individual item sold. `transaction_qty` represents the number of units in a single transaction.
Avg Order Value (AOV)	Average revenue per transaction: `Total Revenue ÷ Total Orders`.
Spending Tier	Classification of transactions by spend amount: Low (≤ R 3.00), Medium (≤ R 6.00), High (> R 6.00).
Day of Week	The name of the day derived from `transaction_date` using `DAYNAME()`.

KPI Definitions

Abbreviation	Full Name	Definition
KPI	Key Performance Indicator	A measurable value that shows how effectively the business is performing.
AOV	Average Order Value	Revenue per transaction: `Total Revenue ÷ Total Orders` = R 4.69.
Revenue	Total Revenue	Sum of `unit_price x transaction_qty` across all transactions = R 698,812.
Total Orders	Total Order Count	Count of distinct `transaction_id` values = 149,116.
Total Units	Total Units Sold	Sum of `transaction_qty` across all transactions.
Distinct Products	Unique Product Count	Count of distinct `product_id` values = 80.
Distinct Stores	Unique Store Count	Count of distinct `store_id` values = 3.

Data Columns

Column	Type	Description
`transaction_id`	integer	Unique identifier for each transaction.
`transaction_date`	datetime	Timestamp of when the transaction occurred. Used for date filtering and time-based charts.
`transaction_time`	datetime	Time component of the transaction (base date 1899-12-31).
`transaction_qty`	integer	Number of units purchased in a single transaction (typically 1 or 2).
`store_id`	integer	Numeric identifier for each store location.
`store_location`	string	Name of the store: Hell's Kitchen, Astoria, or Lower Manhattan.
`product_id`	integer	Unique numeric identifier for each product.
`unit_price`	string	Price per unit, stored as a string with comma decimal separators (e.g., `"3,00"` = R 3.00).
`product_category`	string	High-level product classification (Coffee, Tea, Bakery, etc.).
`product_type`	string	Sub-classification within a category (e.g., Latte, Cappuccino, Espresso).
`product_detail`	string	Full product name including size (e.g., "Sustainably Grown Organic Lg").

Derived Dimensions

Term	Definition
Transaction Day	`DATE(transaction_date)` — date portion only. Used for daily trend chart.
Transaction Month	`DATE_TRUNC('MONTH', transaction_date)` — first day of the transaction's month. Used for monthly trend.
Day of Week	`DAYNAME(transaction_date)` — day name (Mon–Sun). Used for order performance chart.
Spending Tier	Derived from `transaction_qty x unit_price`: Low (≤ R 3.00), Medium (≤ R 6.00), High (> R 6.00).

Abbreviations

Abbreviation	Meaning
ZAR	South African Rand — the currency used for all monetary values (symbol: R).
KPI	Key Performance Indicator.
AOV	Average Order Value.
Lg	Large — product size designation in `product_detail`.
Rg	Regular — product size designation in `product_detail`.
Sm	Small — product size designation in `product_detail`.
POS	Point of Sale — the system where transactions are recorded.
SQL	Structured Query Language — used to query the underlying data.
MEASURE()	SQL function to reference a pre-defined measure in a metric view.

Technical Terms

Term	Definition
Metric View	A semantic layer object defining measures (aggregations) and dimensions separately, enabling flexible grouping at query time.
Measure	A pre-defined aggregation in a metric view (e.g., `revenue`, `total_orders`). Referenced using `MEASURE()`.
Dimension	A non-aggregated field used for grouping and filtering (e.g., `product_category`, `store_location`).
Materialisation	Pre-computation of dataset results for faster dashboard performance when published.
Disaggregated Data	Raw, ungrouped row-level data. The original table has 149,116 rows.
Page-Level Filter	A filter on a canvas page affecting all widgets on that page. All filters on the Sales Performance page are page-level.
Multi-Select Filter	A dropdown allowing selection of one or more values. All values selected by default; deselecting excludes values.